C - Program →

```
# include
int main()
{
int a, b, sum;
Printf("Enter two number:");
Scanf("%d %d", &a; &b;);

Sum = a+b;
Printf("Sum = %d", Sum);

return 0;

};
```

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# Assignment - (2)

## C Programming

Ans (1) Explain the basic structure of a (C) program?

Sol Explaination →

A C programming is generally written in a fixed logical structure. The main parts are :→

1. Documentation / Comments :→ Used to describe the program.
2. Preprocessor Directive :→ Header files are includes using #include
3. Variable Declaration :→ Variables are declared before they are used.
4. Main() functions :→ Execution of a C program starts from the main function.
5. Statements / Program logic :→ Input, processing, calculations and desisions are written here.
6. Return Statement :→ Return(0), normally indicates sucessful completion of main().

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C - program →

```
# include
int main()
{
int n, i, flag=0;
Printf("Enter a number:");
Scanf("%d", &n;);
if (n<=1)
flag=0
else
{
for(i=2; i*i<=n; i++)
{
if (n%.i==0)
{
flag=1
break;
}
}
}
```

→

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Ques (2) Write a C program to check whether a given number is prime or not?

Sol→ Explaination → A prime number greater than (1)

That has exactly two factors (1) and the number itself. The program check whether the given number is divisible by any number from (2) up to its square roots.

```
→ if (flag == 0)
Printf ("%.d is a prime number.", n);
else
Printf("%.d is not a prime number." n);
return 0;
}
```

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C - program →

```
# include
int main()
{
int a,b;

Printf ("Enter two number;");
Scanf (" %d, %d", &d; &b;);

if (b!=0)
 {
if (a%.b==0)
printf ("%d is divisible by %d", a,b);
else
printf ("%d is not divisible by %d", a,b);
}

else

printf ("Division by zero is not allowed.");

return 0;

}
```

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Ques (3) Write a C program to check wheather a given number is divisible or not?

Explanation →

Here, "Divisible to taken to mean checking whether on given number is exactly divisible by another given number. If the remainder by the second number.

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C-programming →

```
#include
  int main()
{
int A[2][2], B[2][2], add[2][2], sub[2][2].
 mul[2][2];
int i, j, k;

printf("Enter elements of matrix A:\n")
for(i=0; i<2; i++)
for (j=0; j<2; j++)
scanf("%d", &A[i][j]);
printf("Enter elements of Matrix B:\n");
for(i=0; i<2; i++)
for (j=0, j<2; j++)
scanf("%d", &B[i][j]);
for (i=0; i<2; i++)
{
for(j=0; j<2; j++)
{
add[i][j] = A[i][j] + B[i][j];
sub[i][j] = A[i][j] - B[i][j];
mul[i][j]=0;
for (K=0; K<2; K++).
mul[0][j] += A[i][k] * B[k][j];
}
}
```

→ - -

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Ques (4) Write a C program to perform matrix addition substraction and multiplication?

Explanation →

for addition and Substraction, both matrixs must have the same the number of row and coloum. for multiplication if matrix (A) is r x c. matrix B must be c x k. The program below uses 2x2 matrixes which is simple to write in an assignment.

--→
```
printf ("\n Addition: \n");
for (i=0; i<2; i++)
printf ("%d %d \n", add[i][0], add[i][1]);

printf ("\n Subtraction: \n");
for (i=0; i<2; i++)
printf ("%d %d \n", Sub[i][0], Sub[i][1]);

printf ("\n Multiplication: \n");
for (i=0; i<2; i++);
printf ("%d %d \n", mul[i][0], mul[i][1];

return 0;
}
```

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## C - program →

```
# include
int main (i)
{
int a, b, c largest;
printf ("Enter three number:");
scanf ("%d %d %d", &a; &b; &c;);

if (a>=b && a>=c)
largest = a;
else if (b>=a && b>=c)
largest = b;
else
largest = c;

printf (" largest number = %d", largest);

return 0;

}
```

Input = 25, 48, 31
output = Largest Number = 48.

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Ques

(5) Write a C program to find the largest among 3 number using conditional statements?

Sol→ Explanation →

The program uses if-else conditional statement. First, it compares the first number with the other two if it is not the largest the second number is compared with remaining numbers. Otherwise the third number is the largest.

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